

EX NO :1	TestNG Annotations – Lifecycle Order Demonstration

AIM:

To demonstrate the usage of TestNG lifecycle annotations like @BeforeSuite, @BeforeTest, @BeforeClass, @BeforeMethod, @Test, and their corresponding @After annotations.

Procedure:

1. Use appropriate TestNG annotations to manage test lifecycle.
2. Observe the order of execution.
3. Create two test methods (testCase1 and testCase2).
4. Run the test suite and verify the output sequence.

PROGRAM:

```
package com.bca;

import org.testng.annotations.*;

public class testcase {

    @BeforeSuite
    public void beforeSuite() {
        System.out.println("Before Suite");
    }

    @AfterSuite
    public void afterSuite() {
        System.out.println("After Suite");
    }

    @BeforeTest
    public void beforeTest() {
        System.out.println("Before Test");
    }

    @AfterTest
    public void afterTest() {
        System.out.println("After Test");
    }

    @BeforeClass
    public void beforeClass() {
        System.out.println("Before Class");
    }

    @AfterClass
    public void afterClass() {
        System.out.println("After Class");
    }

    @BeforeMethod
    public void beforeMethod() {
        System.out.println("Before Each Test Method");
    }

    @AfterMethod
    public void afterMethod() {
        System.out.println("After Each Test Method");
    }
}
```

```
@Test
public void testCase1() {
    System.out.println("Executing Test Case 1");
}

@Test
public void testCase2() {
    System.out.println("Executing Test Case 2");
}
}
```

OUTPUT:

```
-----
Before Suite
Before Test
Before Class
Before Each Test Method
Executing Test Case 1
After Each Test Method
Before Each Test Method
Executing Test Case 2
After Each Test Method
After Class
After Test
PASSED: com.bca.testcase.testCase2
PASSED: com.bca.testcase.testCase1

=====
    Default test
    Tests run: 2, Failures: 0, Skips: 0
=====

After Suite

=====
Default suite
Total tests run: 2, Passes: 2, Failures: 0, Skips: 0
=====
```

Result:

TestNG lifecycle annotations executed in correct order, confirming how the testing framework handles setup and teardown for test execution.

EX NO :2	Chrome Browser using Selenium WebDriver

AIM :

To automate the launching of Chrome browser and open a webpage using Selenium WebDriver.

Procedure:

1. Set system property for Edge driver
2. Initialize EdgeDriver
3. Navigate to <https://www.google.com>
4. Fetch and display the page title

PROGRAM:

```
package selenium;

import org.openqa.selenium.WebDriver;
import org.openqa.selenium.Chrome.ChromeDriver;

public class Test {
    public static void main(String[] args) {

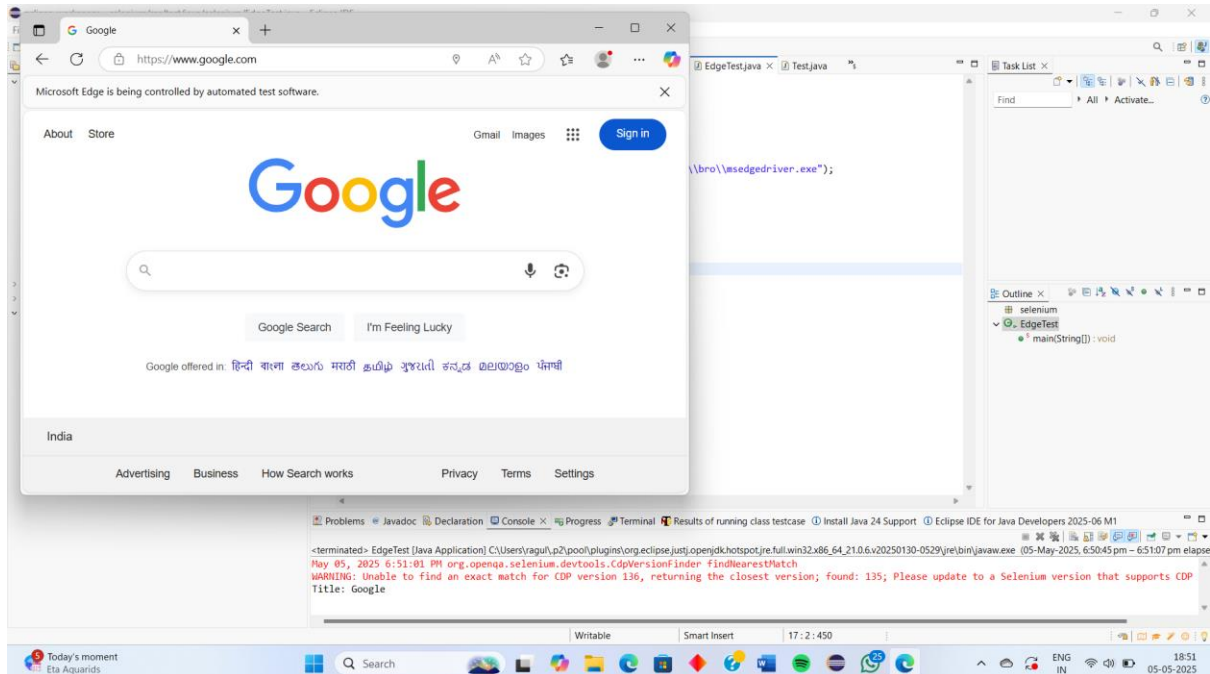
        System.setProperty("webdriver.chrome.driver", "___Path___");

        WebDriver driver = new ChromeDriver();
        driver.get("https://www.google.com");

        System.out.println("Title: " + driver.getTitle());

    }
}
```

OUTPUT :



Result:

Chrome browser launched successfully and title was printed using Selenium WebDriver.

EX NO : 3	TestNG – Priority and Grouping

Aim:

To demonstrate use of priority and groups in TestNG.

Procedure:

1. Create a Java class in a TestNG-enabled project.
2. Use `@Test(priority = x)` on methods, where x is an integer (lower number runs first).
3. Write multiple `@Test` methods with different priorities.
4. Run the program and observe the order of execution in the console.

Program:

```
package com.bca;
```

```
import org.testng.annotations.Test;
```

```
public class EX1 {
```

```
    @Test(priority=2, groups="smoke")
```

```
    public void logoutTest() {
```

```
        System.out.println("Logout Test");
```

```
    }
```

```
    @Test(priority=1, groups="regression")
```

```
    public void loginTest() {
```

```
        System.out.println("Login Test");
```

```
    }
```

```
}
```

OUTPUT :

```
Login Test
Logout Test
PASSED: com.bca.EX1.loginTest
PASSED: com.bca.EX1.logoutTest
```

```
=====
    Default test
    Tests run: 2, Failures: 0, Skips: 0
=====
```

```
=====
Default suite
Total tests run: 2, Passes: 2, Failures: 0, Skips: 0
=====
```

Result:

loginTest runs before logoutTest. You can run by groups via XML.

EX NO : 5	Login Automation using Selenium WebDriver and TestNG

Aim:

To automate login functionality and validate page title after successful login.

Procedure:

1. Open browser
2. Navigate to login page
3. Enter credentials
4. Click login
5. Assert page title

Program :

```
package com.bca;

import org.openqa.selenium.By;
import org.testng.annotations.Test;
import org.openqa.selenium.WebDriver;

import org.openqa.selenium.edge.EdgeDriver;

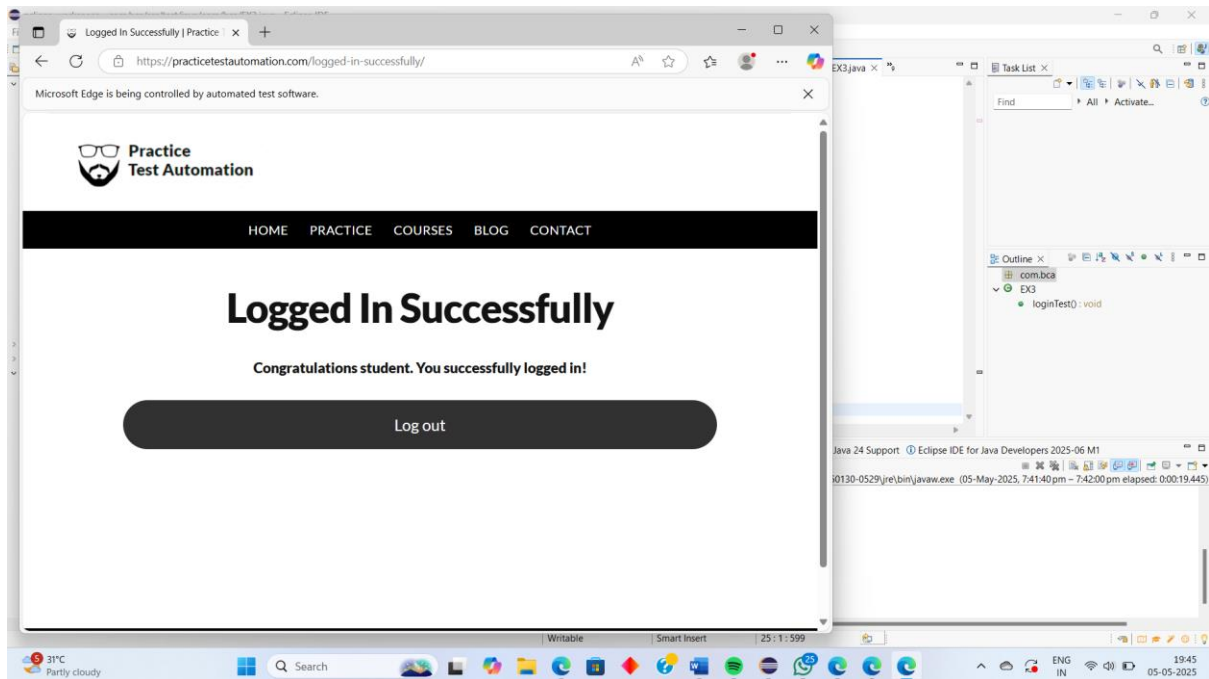
public class EX3 {

    @Test
    public void loginTest() {
        WebDriver driver = new EdgeDriver();
        driver.get("https://practicetestautomation.com/practice-test-
login/");
        driver.findElement(By.id("username")).sendKeys("student");

        driver.findElement(By.id("password")).sendKeys("Password123");
        driver.findElement(By.id("submit")).click();
        driver.getTitle();
        driver.quit();
    }

}
```

OUTPUT :



PASSED: com.bca.EX3.loginTest

Default test

Tests run: 1, Failures: 0, Skips: 0

Default suite

Total tests run: 1, Passes: 1, Failures: 0, Skips: 0

Result:

Login successful and title validated.